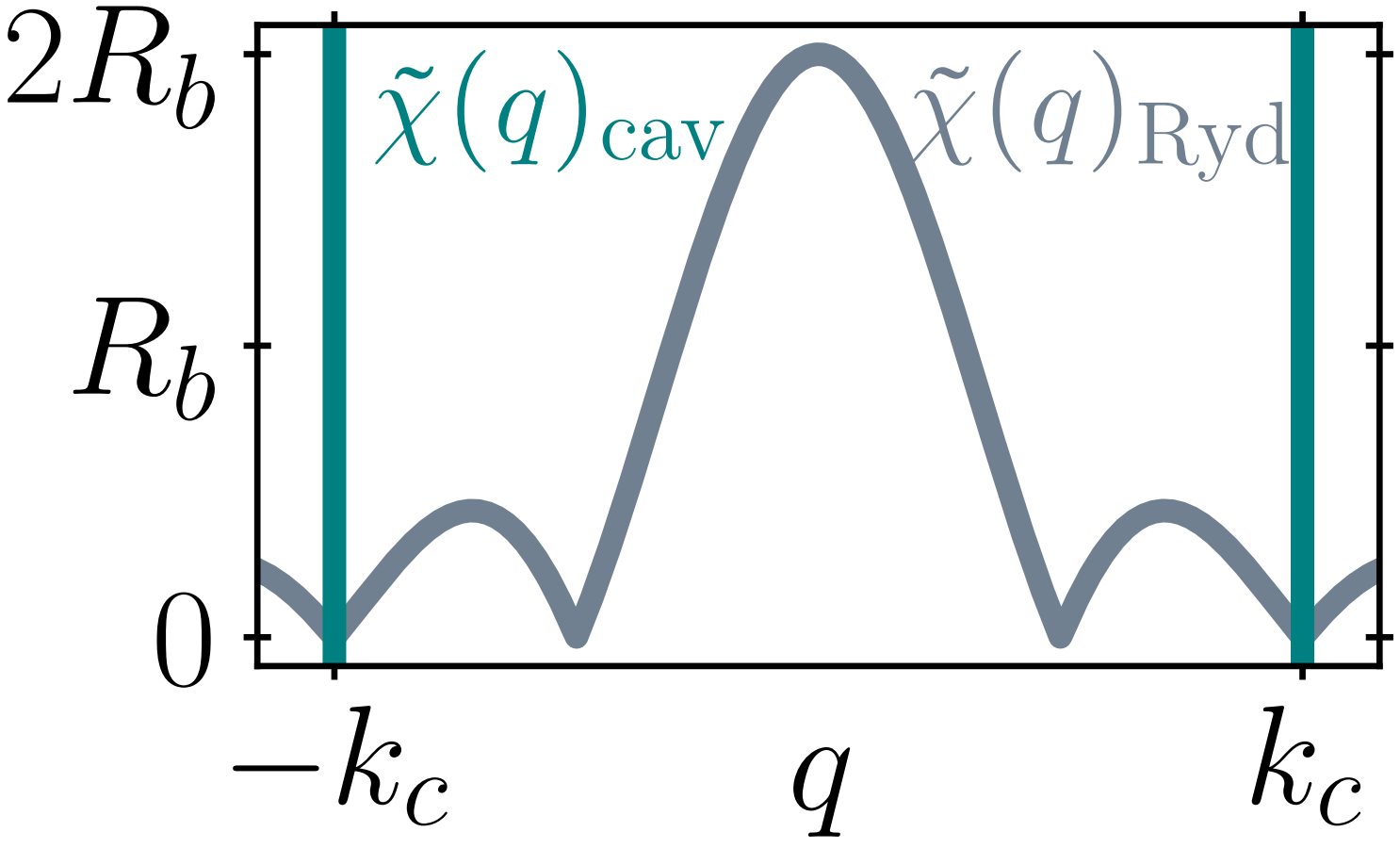
$2R_b$
R_b
0
$\tilde{\chi}(q)_{\mathrm{cav}}$
$\tilde{\chi}(q)_{\mathrm{Ryd}}$
$-k_c$
q
k_c